

PHASE 1 OF 8

# Project Description & Objectives

What LegalEase is, the problem it addresses, and the technologies that make up the solution.

- Overview
- Objectives
- Technologies
- Design

## 1.1 Project Description

LegalEase is an **AI-powered legal document drafting application** designed to simplify the preparation of structured legal document drafts. It accepts the document type, parties, effective date, jurisdiction, terms and any additional instructions, sends the request to a **FastAPI** backend, generates a draft using **Gemini**, and allows the user to review, edit and export the result.

The application is deliberately built as a *drafting assistant*, not a decision-maker: every draft is shown in an editable form so that the user stays in control of the final wording before anything is exported.

## 1.2 Objectives

| Objective                   | What it means in practice                                                           |
|-----------------------------|-------------------------------------------------------------------------------------|
| Generate structured drafts  | Turn a handful of form fields into a complete, numbered legal document draft.       |
| Provide an editable preview | Show the AI draft beside an editable copy so corrections can be made before export. |
| Support TXT, DOCX and PDF   | Let the user download the reviewed document in the format they need.                |
| Separate responsibilities   | Keep frontend, backend, AI generation and export logic in distinct modules.         |

## 1.3 Technologies Used

| Technology            | Role          | Implementation                        |
|-----------------------|---------------|---------------------------------------|
| Python                | Core language | Application logic and services        |
| Streamlit             | Frontend      | Input, preview, editing and downloads |
| FastAPI               | Backend       | REST API and /generate route          |
| Gemini / google-genai | AI            | Legal draft generation                |
| python-docx           | DOCX          | Word document creation                |
| ReportLab             | PDF           | PDF creation                          |
| python-dotenv         | Configuration | Environment variables                 |

## 1.4 The Problem LegalEase Addresses

Preparing even a routine agreement — a freelance contract, an NDA, a rental draft — usually means starting from a blank page or adapting a template that does not quite fit. The work is repetitive: the same clauses (scope, payment, confidentiality, termination, governing law) must be re-worded around new parties, dates and terms. LegalEase reduces that effort. The user supplies the facts once, in plain language; the system produces a well-structured first draft with numbered sections and signature blocks that a person can then refine and, ideally, have reviewed by a qualified legal professional.

## 1.5 Key Capabilities

| Capability            | Description                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------|
| Structured input form | Document type, parties, effective date, jurisdiction, terms and optional instructions.     |
| AI-generated draft    | Gemini produces a professional draft using only the information the user supplied.         |
| Side-by-side review   | A read-only “Generated Document” preview next to an “Editable Version” text area.          |
| Multi-format export   | TXT, DOCX and PDF downloads prepared from the edited text.                                 |
| Built-in disclaimer   | A visible notice reminds users that drafts are informational and need professional review. |

## 1.6 Design Principles

- **Separation of concerns** — the Streamlit interface never talks to Gemini directly; it calls the FastAPI backend, which owns AI access.
- **Human in the loop** — the draft is always editable, and nothing is exported until the user is ready.
- **Secrets stay private** — the Gemini API key lives in a local `.env` file and is never shown in the interface or documentation.
- **Privacy by default** — the application footer states that no generated document is automatically saved by the local application.
- **Extensible by design** — new document types need new input and prompt content, not a new architecture.

## 2. Scenarios

LegalEase is intended for everyday agreements where a structured first draft saves time. Three representative scenarios are described below.

| Scenario                | What the user supplies                                          | What LegalEase produces                         |
|-------------------------|-----------------------------------------------------------------|-------------------------------------------------|
| 1 - Freelance Contract  | Project, payment, confidentiality and termination requirements. | A structured freelance work contract.           |
| 2 - NDA                 | Parties, confidentiality scope and effective date.              | A structured non-disclosure agreement draft.    |
| 3 - Lease or Employment | Document type plus the relevant parties, dates and terms.       | A structured lease or employment-related draft. |

### 2.1 Demonstrated Test Scenario

The demonstrated workflow used a **Freelance Work Contract** between **Arun Kumar** (Freelancer) and **BrightTech Solutions** (Client), effective **25 September 2026**, with **Tamil Nadu, India** as the jurisdiction.

The terms entered were:

- Website-development services;
- ₹50,000 total payment — 50% upfront and 50% on completion;
- A 30-day project duration;
- Confidentiality obligations for both parties;
- 15 days' written notice for termination.

#### Why this scenario matters

It exercises every input field, produces a multi-clause contract with payment stages, and is used throughout this documentation set — in the interface screenshots, the export demonstration and the test plan.

### 2.2 How This Documentation Set Is Organized

| Phase | Title                                 | Covers                                      |
|-------|---------------------------------------|---------------------------------------------|
| 1     | Project Description & Objectives      | Purpose, technologies, scenarios            |
| 2     | Prerequisites, Setup & Architecture   | Setup commands, workflow, project structure |
| 3     | System Architecture & Core API        | Configuration, FastAPI app, schema, route   |
| 4     | AI Integration with Gemini            | Gemini client, prompt requirements          |
| 5     | Frontend, Preview & Editing           | Streamlit UI, input fields, editable draft  |
| 6     | Document Export                       | TXT, DOCX and PDF generation                |
| 7     | Local Deployment & Testing            | Launch, test plan, results                  |
| 8     | Evaluation, Conclusion & Future Scope | Pros, limits, appendix, security            |